Cover Letter to Reviewer 2

Minor Revision Round

Manuscript: "Content-Conditioned Invariant Feature-Statistic Augmentation: Unifying Robustness and Weight Averaging for Domain Generalization" (CIFA)
Journal: *Electronics* (MDPI) **Manuscript ID:** electronics-4654208
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Dear Reviewer 2,

We thank you for the candid and careful assessment, and for acknowledging that the presentation issues have been addressed, that the claims have been appropriately revised, and that the null and non-replicated results are reported transparently. The remaining concerns concern the interpretation and positioning of the work rather than additional experiments. We respond to each point below; the location tag at the start of every response gives the exact page and line range in the tracked PDF, which uses continuous line numbering.

Point 1 — Necessity of the distinctive components

[Tracked PDF: Page 33, Lines 1070–1074; Proposition 1, Pages 8–9, Lines 304–348]

We agree that the content–style decomposition is not a source of measured accuracy: full-moment mixing is statistically indistinguishable from full CIFA (paired $+0.23 \pm 0.75$, $p = 0.53$), and removing weight averaging is the dominant ablation (2.42 points). We would, however, distinguish accuracy contribution from theoretical role. The decomposition is not intended to improve accuracy; it is the structural condition that makes the content-preservation bound admissible. By mixing only the style deviation and fixing the content baseline, the construction guarantees $\mathbb{E}[\mathbf{d} \mid Y] = 0$ and thereby licenses the total-variation bound of Proposition 1. Mixing the full moments does not satisfy this condition and cannot carry the same guarantee, even when its accuracy is matched. The two constructions are tied in accuracy but not in the guarantees they provide; empirical equivalence in accuracy does not establish logical equivalence in the conditions each satisfies.

Point 2 — Value of the unification

[Tracked PDF: Pages 8–11, Lines 304–400]

We explicitly position the contribution as integration rather than a new operator. The unification is not a concatenation of existing techniques: it places feature-moment augmentation, convex-hull minimax, and weight averaging under one causal-invariance principle and states the conditions under which each applies. This yields testable predictions—content–style separability as an enabling rather than sufficient condition, the behavior of the coverage gap, and the second-order flat-minima approximation. Together with the reproducible platform and the falsifiable boundary, this allows later methods to be compared on a common footing and avoids repeated investment in directions already shown to be ineffective. We do not claim a deployment advantage; the practical significance lies in the unified understanding and the well-characterized baseline.

Point 3 — Strength of the assumptions and the coverage term

[Tracked PDF: Page 8, Lines 304–322; Page 36, Lines 1171–1173]

The assumptions are now stated, tested, and falsifiable premises rather than hidden conditions. Content–style separability is labeled an enabling premise, and the trajectory–augmentation alignment is presented as an empirically supported assumption rather than a theorem, with its limitations stated. The target access required to evaluate the coverage-gap term is an inherent property of target-free generalization, not a limitation unique to this work; we demonstrate the behavior of both terms in a controlled moment-space setting (Figure 2, Page 14) rather than presenting the term as computable without target data. A strong assumption is not an incorrect one; the value of the analysis is that its conditions are explicit and open to falsification.

Point 4 — Weakening claims versus adding evidence

[Tracked PDF: Pages 21–23, Lines 795–828; Pages 8–11, Lines 304–400]

When the evidence does not support a stronger claim, weakening that claim is the scientifically correct response; retaining an unsupported claim would be the error. The revision also adds positive evidence—an independent unified-code replication, a gate-pressure stress test, the alignment verification, and a corrected two-one-sided equivalence analysis. This evidence points to statistical equivalence and to boundary conditions rather than to superiority. What was weakened were the claims that did not hold; the claims that do hold—the unifying framework and the boundary characterization—are retained, strengthened, and made more precise and more readily falsifiable.

Point 5 — Overall scientific significance

[Tracked PDF: Page 36, Lines 1168–1200]

The manuscript does not claim a performance breakthrough. Its scientific significance is at the level of understanding: a unifying explanation, a systematic validity boundary, and a falsifiable, reproducible benchmark; the transparent reporting of null and non-replicated results is itself part of that contribution. The only actionable editorial item has been completed (Point 6). We respectfully ask that the work be judged on what it actually establishes rather than on the overstated claims of the original submission.

Point 6 — Editorial cleanup (duplicated or partially edited text)

[Tracked PDF: Page 37, Lines 1209–1211]

We carried out a full editorial review. The one concrete duplication was the data-availability statement, which appeared twice: a manually typeset "**Data availability.**" paragraph and the MDPI template `\dataavailability{}` command with identical content. The manual paragraph has been removed and the template command retained, so the statement now renders exactly once. The remaining items that can look like duplication in a text extraction (redefined revision macros, deliberate indentation and table alignment) were inspected individually and are not duplicated content.

We thank you once again for the thorough and transparent review.

Sincerely,
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